

Hướng dẫn tạo project React + JavaScript + Bootstrap

1. Khởi tạo project với Vite

`npm create vite@latest se1949-se0000-m2 -- --template react`

2. Cài đặt Bootstrap

`npm install react-bootstrap bootstrap`

4. Import Bootstrap CSS

Mở file `src/index.css` và thay toàn bộ nội dung bằng:

```
@import 'bootstrap/dist/css/bootstrap.css';
```

5. Hiện thị Bootstrap CSS

Mở `src/App.jsx` và thử dùng Bootstrap

```
function App() {
  return (
    <>
    <Container className="mt-5">
      <Alert variant="success">
        🎉 Chúc mừng! Bạn đã tích hợp thành công Bootstrap vào React.
      </Alert>
      <Alert variant="success">Đây là thông báo thành công!</Alert>
      <Alert variant="danger">Đã có lỗi xảy ra, vui lòng thử lại.</Alert>
      <Alert variant="warning">Cảnh báo: Pin yếu!</Alert>
      <Alert variant="info">Thông tin: Hệ thống sẽ bảo trì vào 12h đêm.</Alert>
      <Button variant="primary" onClick={() => setCount((count) => count + 1)}>Click thử xem!</Button>

      <Dropdown>
        <Dropdown.Toggle variant="success" id="dropdown-basic">
          Chọn hành động
        </Dropdown.Toggle>

        <Dropdown.Menu>
          <Dropdown.Item href="#/action-1">Chỉnh sửa</Dropdown.Item>
          <Dropdown.Item href="#/action-2">Cài đặt</Dropdown.Item>
          <Dropdown.Divider />
          <Dropdown.Item href="#/action-3" className="text-danger">
            Xóa tài khoản
          </Dropdown.Item>
        </Dropdown.Menu>
      </Dropdown>
    </Container>
    </>
  )
}
```

```

        </Dropdown.Item>
      </Dropdown.Menu>
    </Dropdown>
  </Container>

</>
)
}

export default App

```

6. Chạy project

npm run dev

7. Kết quả

8. Những Component cần quan tâm :

- Table/Grid/Form

```

<Form>
  <Form.Group as={Row} className="mb-3">
    <Form.Label column sm="4">
      Email
    </Form.Label>
    <Col sm="8">
      <Form.Control type="text" value={email} onChange={ (e) => setEmail(e.target.value)} />
    </Col>
  </Form.Group>

  <Form.Group as={Row} className="mb-3">
    <Form.Label column sm="2">
      Password
    </Form.Label>
    <Col sm="10">
      <Form.Control type="password" placeholder="Password" />
    </Col>
  </Form.Group>
</Form>

```

- Label

- TextBox

- Password

- Radio Button (name trùng nhau hết và chỉ khác giá trị/ Chỉ chọn được 1 thôi)

- CheckBox (Được chọn nhiều)

- DropList/Selection

- Button

- Link

- Modal Popup

- TextArea

```
return (
  <div style={{ minHeight: '100vh', backgroundColor: '#f8f9fa' }}>
    { /* Header */ }
    <Navbar bg="light" className="border-bottom">
      <Container fluid>
        <Navbar.Brand>
          <strong>Logo</strong>
        </Navbar.Brand>
        <Navbar.Text className="mx-auto">
          <h4 className="mb-0">Inventory Management</h4>
        </Navbar.Text>
        <Navbar.Text>
          <strong>Date: yyyy-MM-dd</strong>
        </Navbar.Text>
      </Container>
    </Navbar>

    { /* Main Content */ }
    <Container className="mt-4">
      <div className="bg-white p-4 border rounded">
        <h3 className="mb-4">Add New Ingredient</h3>

        <Form>
          { /* Ingredient Name */ }
          <Form.Group as={Row} className="mb-3">
            <Form.Label column sm={2} className="text-end">
              <strong>Ingredient name:</strong>
            </Form.Label>
            <Col sm={10}>
              <Form.Control type="text" />
            </Col>
          </Form.Group>

          { /* Price Range */ }
          <Form.Group as={Row} className="mb-3">
            <Form.Label column sm={3} className="text-end">
              <strong>Price from (d):</strong>
            </Form.Label>
            <Col sm={3}>
              <Form.Control type="number" />
            </Col>
            <Col sm={1} className="text-center">
              <strong>To:</strong>
            </Col>
            <Col sm={3}>
              <Form.Control type="number" />
            </Col>
          </Form.Group>

          { /* Supplier */ }
          <Form.Group as={Row} className="mb-3">
            <Form.Label column sm={3} className="text-end">
              <strong>Supplier:</strong>
            </Form.Label>
            <Col sm={9}>
              <Form.Control type="text" />
            </Col>
          </Form.Group>

          { /* Producer Name */ }
          <Form.Group as={Row} className="mb-3">
            <Form.Label column sm={3} className="text-end">
              <strong>Producer name:</strong>
            </Form.Label>
            <Col sm={9}>
              <Form.Control type="text" />
            </Col>
          </Form.Group>
        </Form>
      </div>
    </Container>
  </div>
)
```

```

    { /* Category and Entry Date */}
    <Form.Group as={Row} className="mb-4">
      <Form.Label column sm={3} className="text-end">
        <strong>Category:</strong>
      </Form.Label>
      <Col sm={3}>
        <Form.Select>
          <option value="">Select category</option>
          <option value="vegetables">Vegetables</option>
          <option value="fruits">Fruits</option>
          <option value="meat">Meat</option>
          <option value="dairy">Dairy</option>
          <option value="spices">Spices</option>
        </Form.Select>
      </Col>
      <Col sm={1}></Col>
      <Form.Label column sm={2} className="text-end">
        <strong>Entry Date (yyyy-MM-dd):</strong>
      </Form.Label>
      <Col sm={3}>
        <Form.Control type="date" />
      </Col>
    </Form.Group>

    { /* Buttons */}
    <Row className="mt-4">
      <Col className="text-center">
        <Button variant="primary" size="lg" className="me-3 px-5">
          Save
        </Button>
        <Button variant="secondary" size="lg" className="px-5">
          Back
        </Button>
      </Col>
    </Row>
  </Form>
</div>
</Container>

{ /* Footer */}
<div className="fixed-bottom bg-light border-top p-3">
  <Container>
    <p className="mb-0">©2026 FPT University</p>
  </Container>
</div>
</div>
);

```

Logo

Inventory Management

Dashboard **Mẫu 1: 2 cột** ...

visualize: Mau 1 two column layout

Add New Ingredient

Ingredient name

Producer name

Price from (đ)

Price to (đ)

Supplier

Unit of measurement

Category

Entry Date (yyyy-MM-dd)

Notes

Save

Back

@2026 FPT University

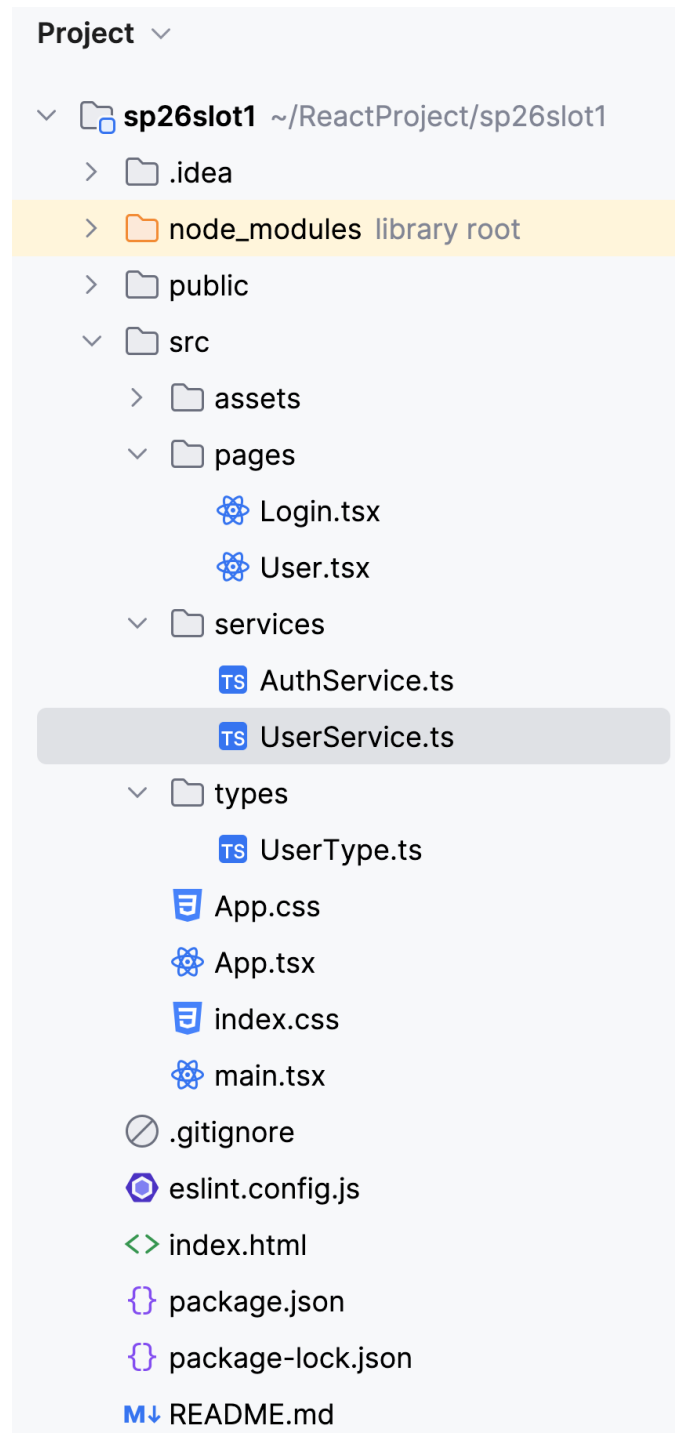


bithub

visualize: Mau 2 three column layout

Basic Info	Pricing	Supplier Info
Ingredient name 	Price from (đ) 	Supplier name
Category 	Price to (đ) 	Producer name
Unit 	Discount (%) 	Contact phone
Entry Date (yyyy-MM-dd) 		Expiry Date (yyyy-MM-dd)
SaveCancel		

Bài 02 : Tạo cấu trúc dự án có front-end



Bài 03 : Tạo Link Route trong

1. Tạo Route

npm install react-router-dom

2. Tạo link ở main.tsx

```
import { StrictMode } from 'react'
import { createRoot } from 'react-dom/client'
import { BrowserRouter, Routes, Route } from 'react-router-dom'
import './index.css'
import App from './App.tsx'
import Login from './pages/Login.tsx'
```

```
createRoot(document.getElementById('root')!).render(
  <StrictMode>
    <BrowserRouter>
      <Routes>
        <Route path="/" element={<App />} />
        <Route path="/login" element={<Login />} />
      </Routes>
    </BrowserRouter>
  </StrictMode>,
)
```

3. Gán link ở trang ReactJS

```
import { Link } from 'react-router-dom'

<Link to="/login" className="text-blue-600 hover:underline font-medium mt-4
inline-block">
  Đăng nhập
</Link>
```



4. Hướng dẫn tích hợp API

npm install axios

Call API

```
import { useState } from "react";
import axios from "axios";
import { useNavigate } from "react-router-dom";

export default function Login() {
  const [email, setEmail] = useState<string>("");
  const [password, setPassword] = useState<string>("");
  const [error, setError] = useState<string>("");
  const [loading, setLoading] = useState<boolean>(false);
  const navigate = useNavigate();

  const handleLogin = async (): Promise<void> => {
    setError("");
    setLoading(true);

    try {
      const response = await axios.post("/api/v1/auth/login", {
        email,
        password,
      });

      localStorage.setItem("token", response.data.token);
      navigate("/products");
    } catch (err) {
      if (axios.isAxiosError(err) && err.response) {
        setError(err.response.data.message || "Email hoặc mật khẩu không đúng");
      } else {
        setError("Không thể kết nối đến server");
      }
    } finally {
      setLoading(false);
    }
  };
};
```

Tách tầng Service

1. Tạo thư viện API chung: src/services/api.js

```
import axios from 'axios';

const api = axios.create({
  baseURL: import.meta.env.VITE_API_BASE_URL ||
'http://localhost:8080/api'
});

export const get = (endpoint) => api.get(endpoint);

export const post = (endpoint, data) => api.post(endpoint, data);

export const put = (endpoint, data) => api.put(endpoint, data);

export const remove = (endpoint) => api.delete(endpoint);

=====

const [lists, setLists] = useState([]);

useEffect(() => {
  const categoryList = async () => {
    try {
      const response = await get("/categories");
      setLists(response.data);
    } catch (error) {
      console.error("Error", error);
    }
  };

  categoryList();
}, []);

const handleDelete = async (id, name) => {
  if (window.confirm('Bạn có chắc muốn xóa ' + name + ' ? ')) {
    try {
      await remove(`/ingredients/${id}`);
      const response = await
get("/ingredients?page=0&size=10&sortBy=ingredientName");
      setIngredientList(response.data.content);
    } catch (error) {
      alert("Lỗi khi xóa!");
    }
  }
};
```

Hướng dẫn thêm và validate

```
const navigate = useNavigate();

const [categoryList, setCategoryList] = useState([]);

const [errors, setErrors] = useState({});

const [formData, setFormData] = useState({
  ingredientName: "",
  priceFrom: 2000,
  priceTo: 3000,
  producerName: "",
  entryDate: "",
  supplier: "",
  categoryId: 0
});

const validate = () => {
  const newErrors = {};
  // Validate tên
  if (!formData.ingredientName) {
    newErrors.ingredientName = 'Tên sản phẩm không  
được để trống';
  } else if (formData.ingredientName.length < 3) {
    newErrors.ingredientName = 'Tên sản phẩm phải có  
ít nhất 3 ký tự';
  }
  // Validate giá
  if (!formData.priceTo) {
    newErrors.priceTo = 'Giá không được để trống';
  } else if (isNaN(formData.priceTo)) {
    newErrors.priceTo = 'Giá phải là số';
  } else if (formData.priceTo <= 0) {
    newErrors.priceTo = 'Giá phải lớn hơn 0';
  }

  return newErrors;
};

const handleChange = (e) => {
  setFormData({
```

```

        ...formData,
        [e.target.name]: e.target.value
    });

    if (errors[name]) {
        setErrors({
            ...errors,
            [name]: ''
        });
    }
};

const handleSubmit = async (e) => {
    e.preventDefault();

    const validationErrors = validate();

    if (Object.keys(validationErrors).length > 0) {
        setErrors(validationErrors);
        return;
    }

    try {
        await post("/ingredients", formData, {
            headers: {
                'Content-Type': 'application/json'
            }
        });
        navigate('/');
    } catch (error) {
        console.log("Error " + error);
    }
};

```

```

<Form.Group as={Row} className="mb-3">
  <Form.Label column sm={3} className="text-end">
    <strong>Ingredient name:</strong>
  </Form.Label>

```

```

    <Col sm={9}>
      <Form.Control type="text" name="ingredientName"
value={formData.ingredientName} onChange={handleChange}
/>
    </Col>
    {errors.ingredientName && <span style={{color:
'red'}}>{errors.ingredientName}</span>}

</Form.Group>

```

2. Sử dụng trong Login.tsx

```

import {authenService} from '../services/AuthenService';

const handleLogin = () => {
  authenService.login(email, password).then((response) => {
    localStorage.setItem("token",
    response.data.accessToken);
    navigate("/app");
  }).catch((err) => {
    console.error(err);
  });
}

```

3. Lấy dữ liệu sử dụng jwt

```

const navigator = useNavigate();
const [users, setUsers] = useState([]);

const userlist = async () => {
  const token = localStorage.getItem("token");
  if (!token) {
    console.error("No token found");
    navigator("/");
    return;
  }
  try {
    const response = await
axios.get("http://localhost:8080/api/v1/users?page=1&size=5",
{
  headers: {
    Authorization: `Bearer ${token}`,
  }
}

```

```

    })
    .then((response) => {
        console.log("ABC", response);
        setUsers(response.data.data.content);
    })
    .catch((error) => {
        console.error(error);
    });
} catch (error) {
    console.error("Error fetching user list:", error);
}
};

useEffect(() => {
    userlist();
}, [])

return (
    <>
        <div>
            <h1>User</h1>
            <table className="user-table">
                <thead>
                    <tr>
                        <th>ID</th>
                        <th>Name</th>
                        <th>Email</th>
                    </tr>
                </thead>
                <tbody>
                    {users.map((user: any) => (
                        <tr key={user.id}>
                            <td>{user.id}</td>
                            <td>{user.email}</td>
                            <td>{user.fullName}</td>
                        </tr>
                    ))}
                </tbody>
            </table>

            <button type="button">Hello</button>
        </div>
    </>

);

```

```
}  
  
export default User;
```

Cấu hình URL bên ngoài

tạo .env :

```
VITE_API_URL=http://localhost:8080
```

Cấu hình API ở Services:

```
const login = async (email: string, password: string) => {  
  const data = await  
  axios.post(`${import.meta.env.VITE_API_URL}/api/v1/auth/login`,  
    {  
      email,  
      password,  
    });  
  return data;  
};
```

